

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY 2021 TRIMESTER

FINAL ASSESSMENT

**UCCD1133 INTRODUCTION TO COMPUTER ORGANISATION AND
ARCHITECTURE**

SATURDAY, 18 SEPTEMBER 2021

TIME : 2.00 PM – 5.00 PM (3 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instructions to Candidates:

REMINDER: You are reminded to read and adhere to the Final Assessment Instructions to Candidates that has been made available through the UTAR Portal before the commencement of Final Assessment (FA). The detailed instructions for this FA are as follows:

General

1. This Final Assessment (FA) is an Individual, time restricted assessment which consists of FOUR (4) questions. Each question carries 25 marks.
2. You are required to answer **ALL** questions, and submit the **ANSWER SCRIPT** by **5.00 pm, 18 SEPTEMBER 2021**.
3. During the period of **3 hours** of this FA, the examiner(s) can be reached at
 - (a). Microsoft Teams, or
 - (b). Email: changjj@utar.edu.myYou may use the above e-platform(s) to check with the examiner(s) if you need any clarification on this FA question paper.
4. You may refer to any books, lecture notes, published materials, online resources etc. when answering the questions. Proper referencing is necessary to avoid plagiarism. The preferred referencing style is American Psychological Association (APA) / Harvard / Chicago Manual. Candidates are reminded that **Copy-and-Paste, Consultation, Discussion and Sharing of Answers** with others are **STRICTLY PROHIBITED** in this FA.

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Answer Script File

5. The answer script **MUST** be a **PDF** file in A4 size format. **Note: Please keep the file size NOT exceeding 30MB per file.**
6. Please check you index number generated by the Division of Examinations, Awards, and Scholarships (DEAS). You **MUST** name your answer script using the following file name for submission:

UCCD1133_FA_[Programme Abbreviation]_[Your Index Number]

For example, if you are from the degree programme CN, and your Index Number is A01234CBCNF, then your answer script should be named as

UCCD1133_FA_CN_A01234CBCNF.docx

Answer Script Submission

7. Your answer script file has to be submitted to the following platform(s) before the due time/date.
 - (a). Attach your answer script at the WBLE submission link
 - (b). Send your answer script to the course code email:
UCCD1133.FICT@utar.edu.my
Note: For the subject / title of your email, use the file name of your answer script, i.e.,
UCCD1133_FA_[Programme Abbreviation]_[Your Index Number]
8. Please make sure you submit the correct, complete and final version of your answer script and the same copy of answer script file is to be submitted to the above platform(s). When multiple answer script files are received, the examiner will randomly choose one for marking.

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9. The **first page** of your submission is the **Final Assessment Cover Page**. You **MUST** use the template given and fill in the following information
 - Your Programme (Abbreviation)
 - Your Index Number
 - Your Name
 - Your Student ID
10. The **second page** of your submission is the **Final Assessment Declaration Statement**. You **MUST** use the template given, and digitally sign on the form to indicate the authenticity of your submitted work is without plagiarism.
11. Each question should be answered starting on a new page. It is recommended that the answer to each question is limited to **5 pages** or **500 words**, whichever is lower.
12. For answer scripts that have text-based answers only, all texts **MUST** be typed and recommended using Times New Roman characters with font size **12**, with proper spacing and alignment, except for the drawings and equations/calculations.
13. For answer scripts that require / contain drawings, equations and calculations with short text descriptions, you can hand-write your answers on a blank paper, and then use the scanner apps in your smartphone (Scanner Document, CamScanner, Adobe Scanner etc.) to take scanned copy, or you can type if necessary and include the scanned copy taken in the Word document, as part of your submitted answers.
14. Please include a page number on **each and every page** of your answer script. Ensure that each page of answer scripts is in sequence prior to online submission.

WARNING OF PLAGIARISM

15. Answer scripts shall be uploaded to Turnitin to check for the originality of submitted answers. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have plagiarized in the answer submitted. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

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Q1. (a) Answer the following questions. Show all your workings.

- (i) Let X be a decimal number equals to the last four digits of your student ID. Convert X to base 14. (3 marks)
- (ii) Compute the decimal value of the binary number 1100 1010 0011 if the given number represents *unsigned integer*. Repeat if the number represents *2's complement* (i.e., compute the decimal value of the same number if it is 2's complement). Assume a 12-bits system. (5 marks)
- (iii) Perform the following arithmetic operation using 2's complement representation: $2D1_{16} - 0000\ 0011\ 0101_{\text{Gray}}$. Then, convert your answer to octal (base 8). Assume a 12-bits system. (6 marks)
- (iv) Simplify the following expression using Boolean algebra

$$F(A, B, C) = (A + B + C)'D + AD + B \quad (4 \text{ marks})$$

- (b) Figure 1.1 shows a combinational circuit consists of a NOT gate, a multiplexer and a full adder.

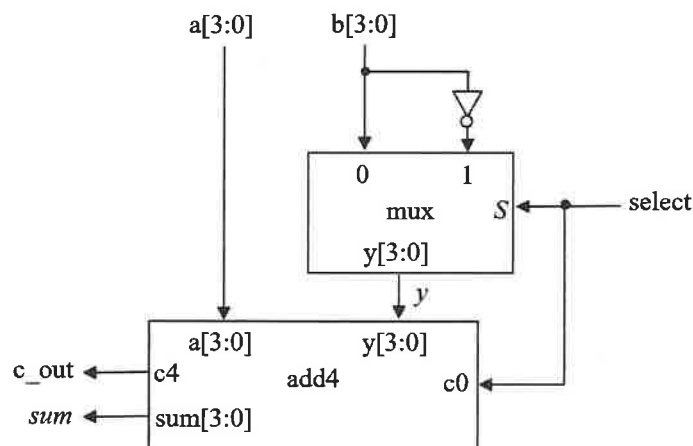


Figure 1.1

- (i) What is the value of y if $a = 0010$, $b = 0110$ and $S = 0$? (1 mark)
- (ii) What is the value of y if $a = 0010$, $b = 0110$ and $S = 1$? (2 marks)
- (iii) What is the value of sum if $a = 0010$, $b = 0110$ and $S = 0$? Show your working. (2 marks)
- (iv) What is the value of sum if $a = 0010$, $b = 0110$ and $S = 1$? Show your working. (2 marks)

[Total : 25 marks]

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Q2. (a) Consider the following machine codes

0x02208020 ... instruction (1)

0xAD300064 ... instruction (2)

- (i) Translate the two machine codes above to MIPS assembly code. Show all your workings. (10 mark)
- (ii) Which pseudoinstruction can be used to replace assembly code for instruction (1)? Rewrite the assembly code with a MIPS pseudoinstruction that do the same task. (2 marks)
- (iii) State the addressing mode of instruction (2). (1 mark)

(b) Consider the following MIPS assembly code.

```

        slt    $t2, $t0, $t1
        beq    $t2, $0, ELSE
        j      DONE
ELSE:    addi   $t2, $0, 3
DONE:
```

Suppose that \$t1 has the value of 0x11223344.

- (i) What is the value of \$t2 after the instructions above if \$t0 = 0x1055FF88? (2 marks)
- (ii) What is the value of \$t2 after the instructions above if \$t0 = 0x22220000? (2 marks)
- (iii) Translate the assembly code “beq \$t2, \$0, ELSE” to machine language in hexadecimal. Show all your workings. (5 marks)
- (iv) The label ELSE is used before it is defined. Why is this permitted? (3 marks)

[Total : 25 marks]

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- Q3. (a) In order to reduce the size of a MIPS processor, Hisham decided to reduce the instruction format size from 32 bits to 24 bits.
- (i) Under the new instruction format, the register fields are only 4-bit long. How many registers could the new machine have? (2 marks)
 - (ii) How does reducing the register-field length affect the processor size? (2 marks)
 - (iii) Suggest ONE (1) other method to reduce the size of the instruction. (2 marks)
- (b) Jane notice that her single cycle processor has a slower clock rate compared to a multicycle processor. She wanted to increase the performance of her processor by increasing the clock rate. Would you advise her to increase the clock rate to be as fast as the multicycle processor? Why? Explain your answer by discussing the differences between single cycle and multi-cycle microarchitectures. (11 marks)
- (c) Discuss TWO (2) similarities and TWO (2) differences between compiler and interpreter. (8 marks)
[Total : 25 marks]
- Q4. (a) (i) State TWO (2) advantages of using direct memory access (DMA) to transfer data between peripherals and main memory. (4 marks)
- (ii) Both the DMA and CPU need to use the memory bus at the same time. What should the DMA controller do? (6 marks)
- (b) A MIPS assembly language programmer accidentally spelt the mnemonics wrongly and caused an error due to undefined instruction. Explain the actions taken by the MIPS processor when such an error occurs. (10 marks)
- (c) SPIM is a software simulator that runs assembly language programs written for processors that implement the MIPS-32 architecture. Why do we say that it uses memory mapped I/O but not isolated I/O? (5 marks)
[Total : 25 marks]